

# Teleportation Service

Dario Bandecchi

Giacomo Raffo



*“Teleportation Service” – Quantum Computing and Quantum Internet*



# Overview

- The objective of this project is to simulate an application-level perspective of a **Quantum Network** formed by  $N$  fully-connected end-nodes (managed by a central virtual "Quantum Network Core") which can request the teleportation of arbitrary qubit states  $\rho$
- The requests are served sequentially
- A depolarization probability  $p_w$  is applied to the Bell pair used for the teleportation
- The simulation is done implementing the teleportation protocol in both **Qiskit/Python** and **QuantumSavory/Julia**
- The goal is to analyze the fidelity between the original state  $\rho$  and the teleported depolarized state  $\rho'$



# Simulation Workflow

- For both the **Qiskit/Python** and **QuantumSavory/Julia** implementations, the simulation parameters are the following:
  - Number of nodes in the network
  - Number of requests per depolarization probability value
  - Amplitudes of the input state
- To perform the simulations, we evaluated 20 different values of depolarization probability. For each value, a predefined number of teleportation requests was executed between randomly selected sender and receiver nodes
- For each request, the fidelity between the input and output states was computed
- To generate the plots, the fidelity values were averaged across all requests performed for each specific error probability level
- We considered only the case of teleportation of pure input states  $|\psi\rangle$



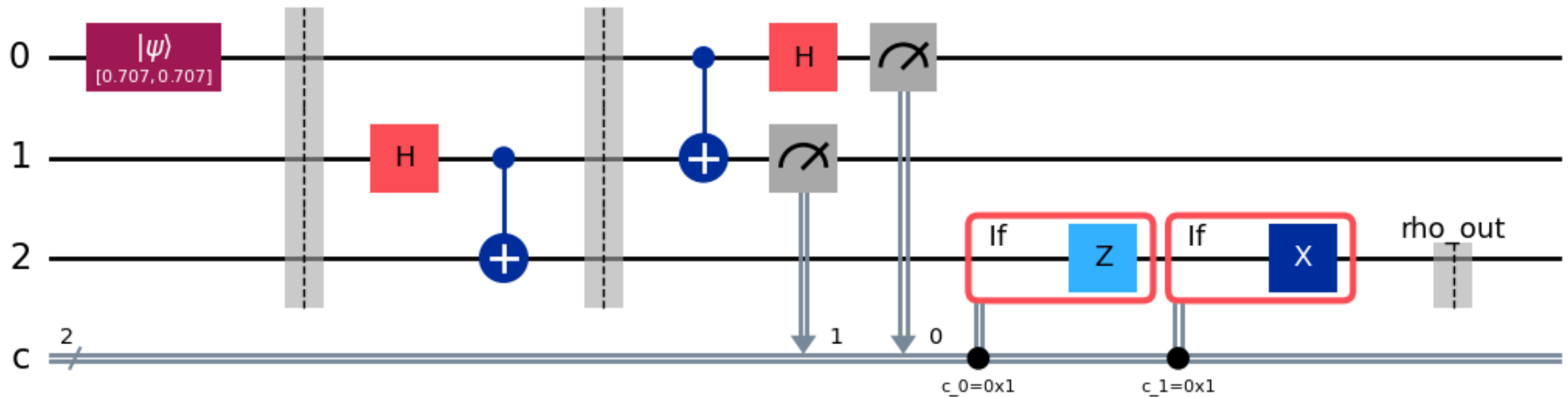
# Qiskit Implementation

- For each request, a new **QuantumCircuit** is created involving three qubits: two for the Sender (the input state and one half of the entangled pair) and one for the Receiver (the second half of the entangled pair)
- To simulate the generation of depolarized Bell pairs, a **NoiseModel** with **depolarizing\_error** is applied to the CNOT gate during the Bell state generation
- The circuits are executed on **AerSimulator** using *method='density\_matrix'* to extract the teleported density matrix  $\rho'$  from the Receiver's register
- The fidelity is calculated through the **state\_fidelity** function, which implements the following formula:

$$F = \langle \psi | \rho' | \psi \rangle$$



# Qiskit Implementation – Teleportation Circuit



# QuantumSavory Implementation

- For each request 3 protocols are activated, one for every agent of the simulation:
  - The **Quantum Network Core Protocol** creates a depolarized Bell pair between the Sender and the Receiver with an exponentially distributed delay of mean equal to *200ms*
  - The **Sender Protocol** waits for the entangled pair to be created, performs a Bell State Measurement on the input state and the Sender's qubit of the entangled pair, and sends the measurement results to the Receiver
  - The **Receiver Protocol** waits for the entangled pair to be created, then waits for the measurement results from the Sender, and finally applies the appropriate corrections to the Receiver's qubit of the entangled pair based on the measurement results
- **Classical Delay:** Measurement results are sent via a classical channel with propagation delay
- **Resource Tagging:** The Core prepares Bell pairs and tags the nodes' registers instantly after the exponential timeout expires



*"Teleportation Service" – Quantum Computing and Quantum Internet*

# QuantumSavory Implementation

- For the fidelity calculation we used the ***observable*** function, which computes the expectation value of a quantum observable on the given register
- In particular:

```
fidelity = observable([net[dst]], [2], rho_in)
```

where the register slot contains the teleported state ( $\rho'$ ) and *rho\_in* acts as observable

- In this case, this approach is mathematically equivalent to the fidelity formula, as detailed in the next slide



# QuantumSavory Implementation

- This can be easily proved considering that the function computes the following expression

$$\langle O \rangle = \text{tr}(O\rho)$$

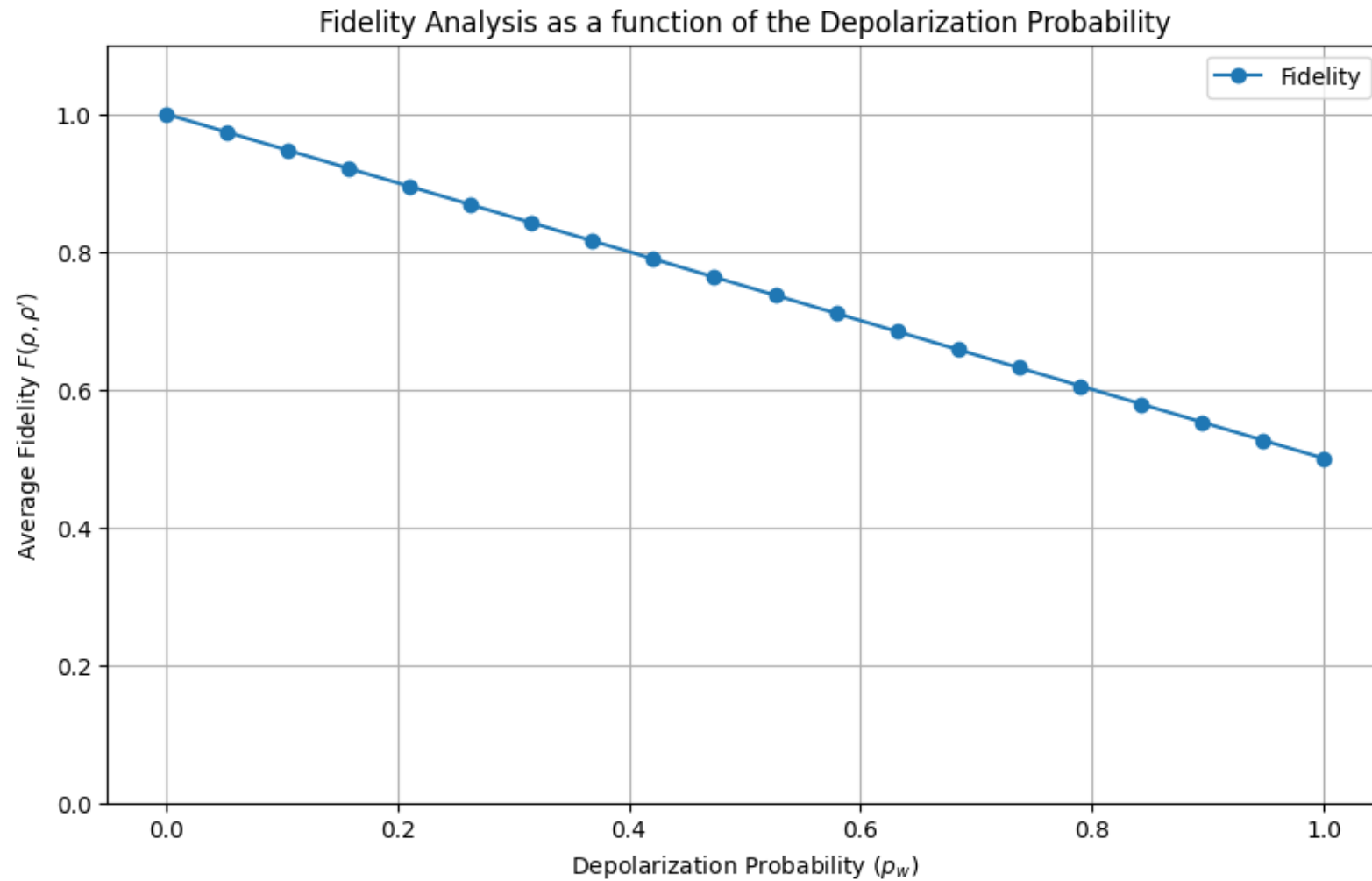
- By noting that in our case the observable  $O$  is equal to  $|\psi\rangle\langle\psi|$ , and substituting the teleported state  $\rho = \rho'$ , we obtain

$$F = \text{tr}(|\psi\rangle\langle\psi|\rho') = \text{tr}(\langle\psi|\rho'|\psi\rangle) = \langle\psi|\rho'|\psi\rangle$$

which is exactly the expected formula for the fidelity

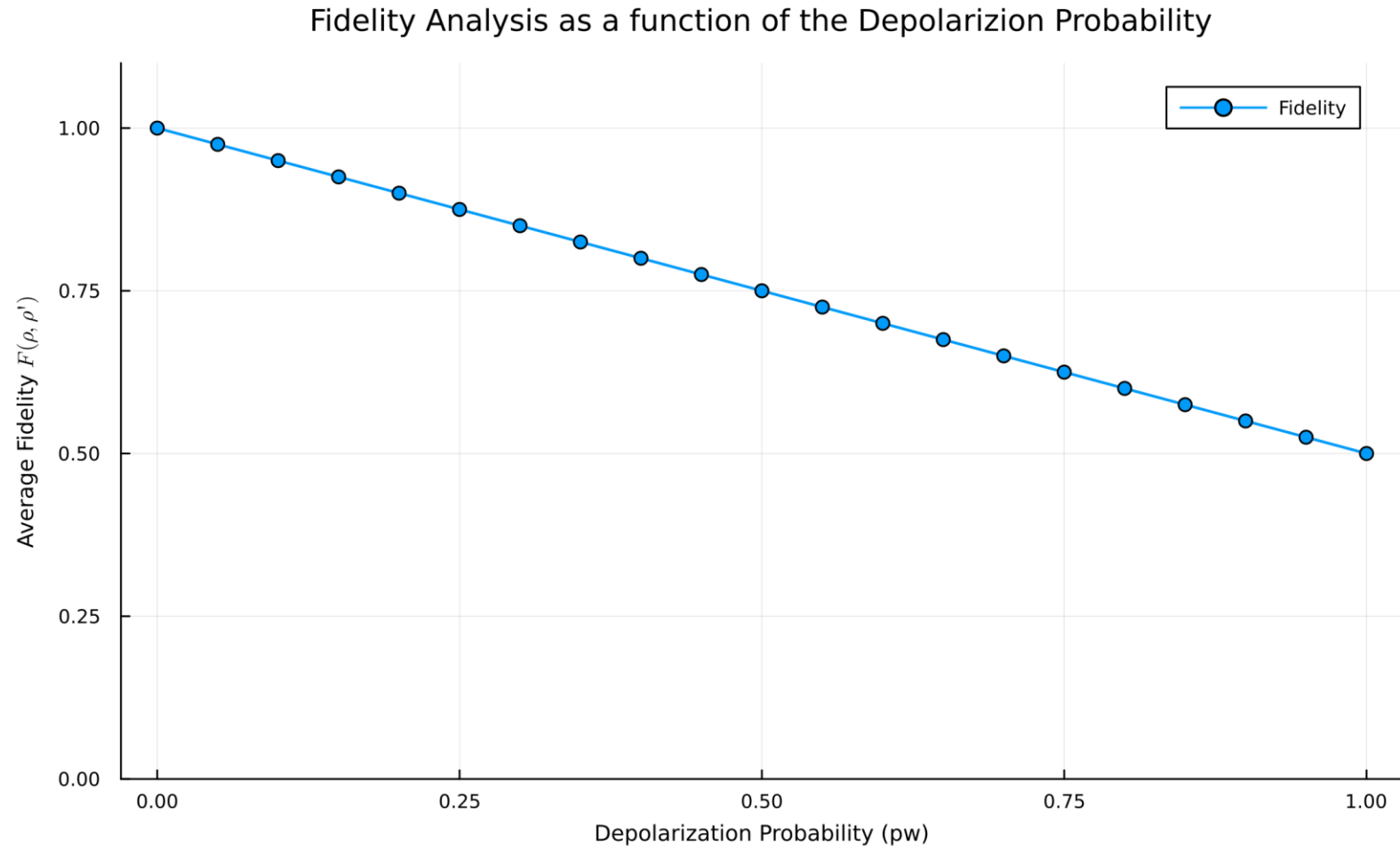


# Fidelity Analysis - Qiskit



*“Teleportation Service” – Quantum Computing and Quantum Internet*

# Fidelity Analysis - QuantumSavory



*“Teleportation Service” – Quantum Computing and Quantum Internet*

# Algebraic Demonstration of the Results

- To demonstrate the graphical results obtained from the simulations, we can start considering the depolarized version of the Bell pair

$$\rho = (1-p_w)|\Phi^+\rangle\langle\Phi^+| + p_w \mathbb{I}/4$$

and by exploiting the completeness relation

$$\mathbb{I} = |\Phi^+\rangle\langle\Phi^+| + |\Phi^-\rangle\langle\Phi^-| + |\Psi^+\rangle\langle\Psi^+| + |\Psi^-\rangle\langle\Psi^-|$$

we derive the following expression

$$\rho = (1-\frac{3}{4}p_w)|\Phi^+\rangle\langle\Phi^+| + \frac{p_w}{4} (|\Phi^-\rangle\langle\Phi^-| + |\Psi^+\rangle\langle\Psi^+| + |\Psi^-\rangle\langle\Psi^-|)$$

- This result matches the standard form of a **Werner State**

$$\rho = F|\Phi^+\rangle\langle\Phi^+| + \frac{1-F}{3} (|\Phi^-\rangle\langle\Phi^-| + |\Psi^+\rangle\langle\Psi^+| + |\Psi^-\rangle\langle\Psi^-|)$$



where the fidelity  $F$  is equal to  $1-\frac{3}{4}p_w$

*“Teleportation Service” – Quantum Computing and Quantum Internet*

# Algebraic Demonstration of the Results

- Now if we use this state to perform the teleportation protocol, the teleported state  $\rho'$  would be affected by depolarizing noise, and then to see how close is to the initial state  $|\psi\rangle$  we can compute the final fidelity

$$\begin{aligned}\hat{F} = \langle\psi|\rho'|\psi\rangle &= \frac{2F + 1}{3} = \frac{2\left(1 - \frac{3}{4}p_w\right) + 1}{3} = \frac{2 - \frac{3}{2}p_w + 1}{3} = \\ &= \frac{3 - \frac{3}{2}p_w}{3} = \boxed{1 - \frac{p_w}{2}}\end{aligned}$$

which is exactly the equation of the line shown in the graphs

- Finally, the results confirm that the fidelity is independent of the chosen input state  $|\psi\rangle$

